

C1 Four Topology MOSbius

Layout Review Slides 08/10

[LINK HERE](#)

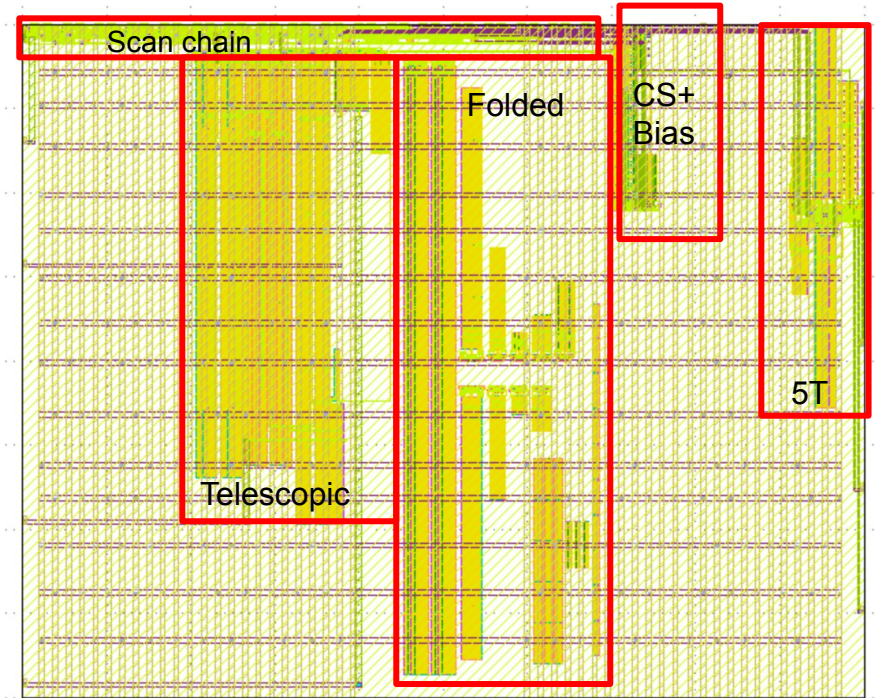
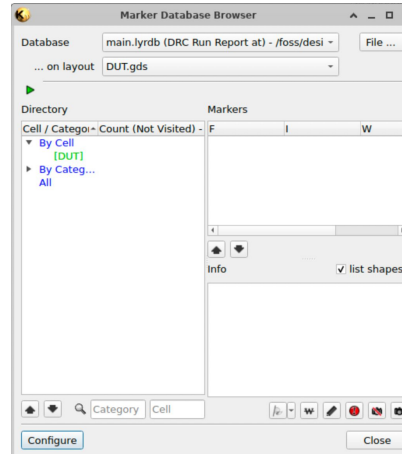
Discord name	Affiliation	Experience	Role
elij0100	Columbia University	Undergraduate	Team Lead
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maxdrucker_83950	Columbia University	Undergraduate	Team Member
Manwell	Columbia University	Undergraduate	Team Member

General Updates

- The most recent GDS is available in the repository
- The top level is currently DRC clean but not yet LVS clean. The Folded Cascode layout is still in progress and needs to be routed into the top level. All other blocks at the top level are DRC and LVS clean.
- We expect to have a fully DRC and LVS clean Top Level layout by Wednesday and will continue to update track leads on our progress

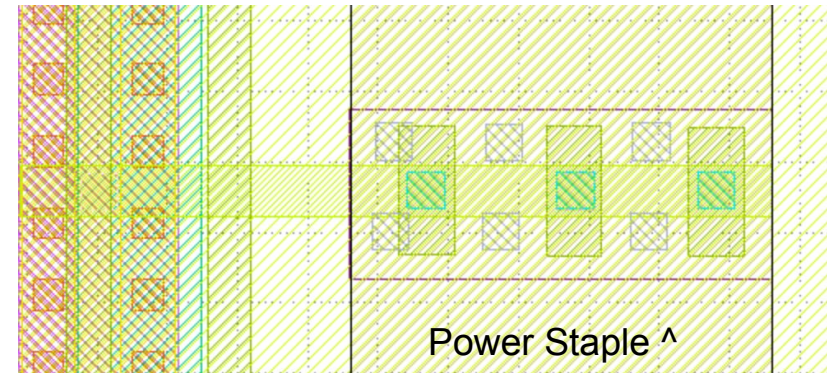
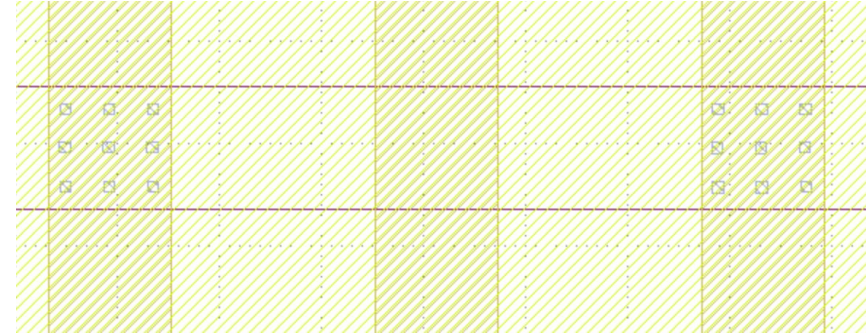
Top Level Integration (Max)

- Top level GDS measures $500\mu\text{m} \times 400\mu\text{m}$
- 18 total pins: 5 on top, 5 on bottom, 4 on left, 4 on right
- Pins are distributed along the edges to simplify wire bonding
- All routing to pins uses wide top-level metal traces
- Top level is DRC clean

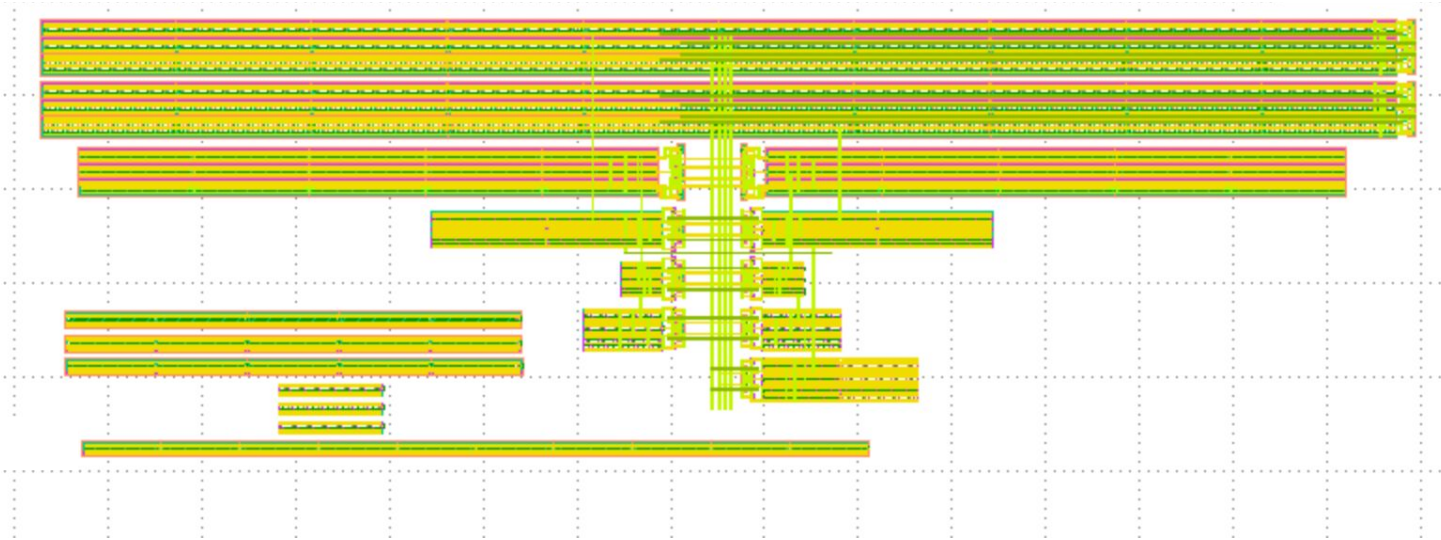


Power Grid (Max)

- Power grid uses $3\mu\text{m}$ thick wires on Metal4 and Metal5
- Overlapping Metal4/Metal5 regions fit 9 vias each, significantly reducing parasitic resistance and ensuring power integrity
- "Power staples" use the same approach to deliver power to blocks within cells
- Grid distributes power evenly across the entire chip
- Overlapping metal layers provide metal-to-metal decoupling capacitance
- Wide wires limit current density to avoid electromigration



Folded Cascode Layout (Eli)



Folded Cascode Layout Considerations

- Matched devices placed with mirror symmetry, all gates running the same direction so both devices see the same process conditions
- Longer L on current mirrors and current sources ($L = 1\mu\text{m}$) for better threshold voltage matching
- Left and right branches placed symmetrically, aside from the PMOS current sources
- $1.4\mu\text{m}+$ gap between N-well and P-sub regions for noise isolation between PMOS and NMOS groups (also a design rule requirement)
- Bulk ties on all transistors, giving a low-impedance path from the body to a clean supply (VDD for PMOS, VSS for NMOS) to reduce substrate noise coupling
- Routing convention: Metal1 horizontal, Metal2 vertical, Metal3 horizontal for enable and bias buses

Folded Cascode Layout Next Steps

- Currently DRC Clean
 - Floor plan complete
 - Left branch and right branch fully routed
- Final To Do (to pass LVS)
 - Bias network routing
 - VSS, VDD connections
 - Make pins for inputs, outputs
- Should be DRC/LVS clean by Wednesday

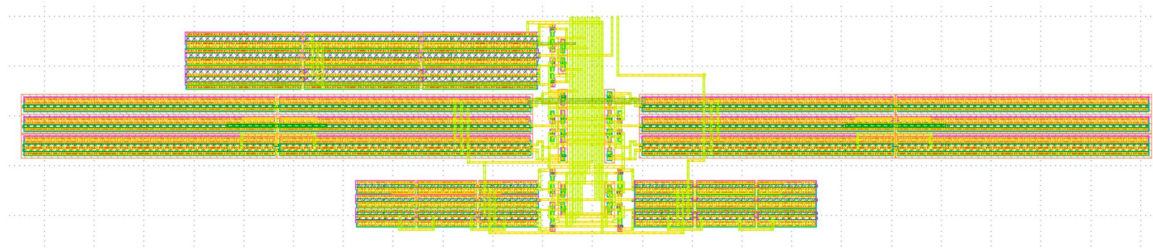
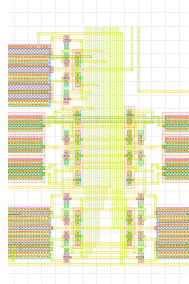
Telescopic Cascode Layout (Manuel + Max)

- Telescopic cascode layout has enable signals routed along the bottom edge
- Matched transistor pairs placed side by side with the same orientation for best matching during fabrication
- Metal2 used for longer routing runs over Metal1 to reduce parasitic resistance
- Bias network connected in the top left corner, grouped with all non-enable analog pins
- Analog signal pins kept physically separated from digital enable pins to minimize noise coupling



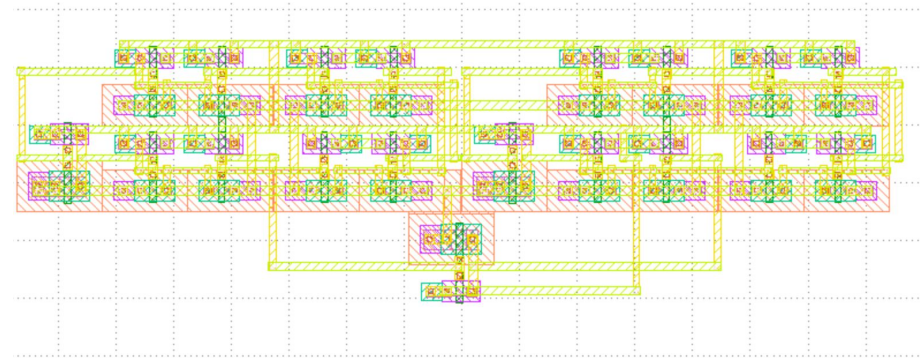
5 Transistor (Songhang)

- Programmable transistors placed face to face to simplify routing and reduce interconnect length
- Enable, input, and bias pins placed along the top edge for clean external connection
- Symmetric placement used to improve matching between devices
- Vertical and centralized routing for an organized layout
- DRC clean, LVS clean
- Cell dimensions: $226\mu\text{m} \times 43\mu\text{m}$



Digital (Max)

- Scan chain consists of 9 D flip-flops
- Each DFF is built from two D latches and a clock inverter
- Metal hopping used on gate connections to avoid antenna effects
- DFFs stacked left to right, aligned with the Metal2 routing direction for efficient long-path connections



Thank You!
Any feedback is greatly appreciated.